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CHAPTER 1: Introduction & Architectural Overview

1.1 Academic Genesis & Professional Motivation

The inception of this project stems from a deep-seated fascination with computer microarchitecture, digital logic design, and the hardware-software interface. This endeavour represents the practical synthesis of theoretical principles acquired through specialized coursework in computer architecture via Coursera, alongside industrial-oriented exposure gained during a microarchitecture and VLSI training internship through Wipro in association with Mood Indigo, IIT Bombay.

Engaging with hardware description at the register-transfer level (RTL) during these programs solidified my passion for semiconductor design, processor pipelining, and high-performance computing architectures. Rather than treating hardware as an abstract black box that merely executes compiled software binaries, this work was driven by the desire to construct a fully functioning, instruction-set-compliant silicon engine from the ground up. I intend to build upon this foundational implementation by actively participating in advanced microarchitectural research, open-source silicon development, and next-generation Out-of-Order (OoO) processor initiatives.

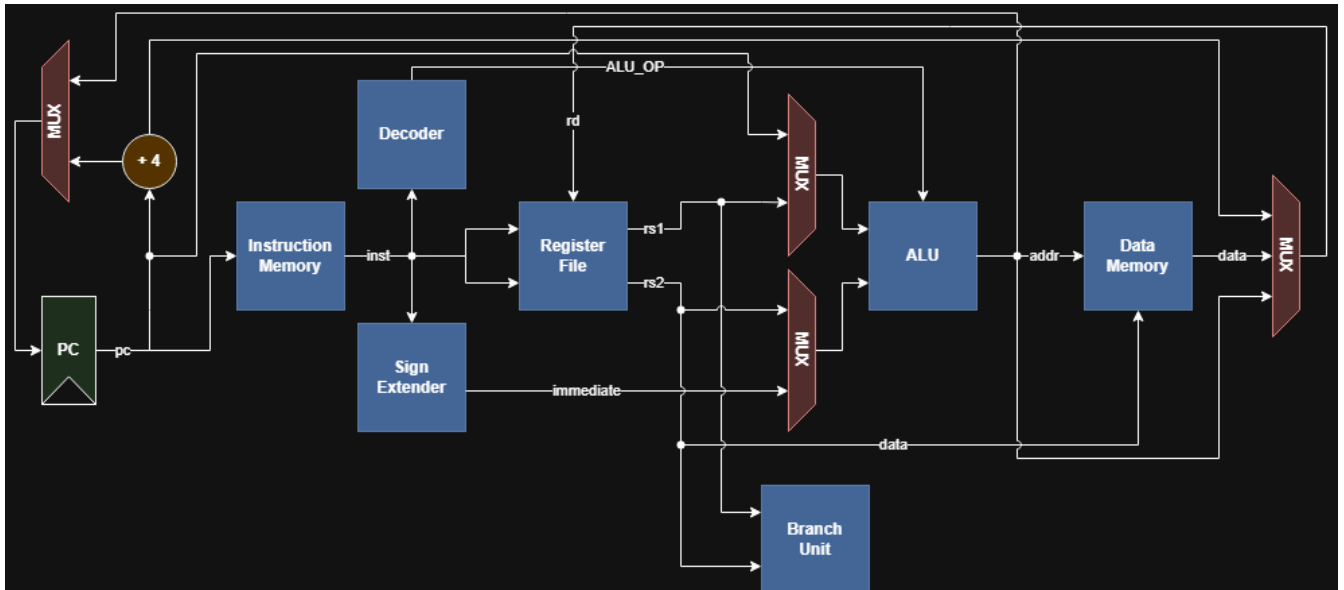
1.2 Background on the RISC-V Open Architecture

Over the past five decades, the computing landscape has been largely dominated by proprietary, closed-source Instruction Set Architectures (ISAs) such as x86 and ARM. While commercially successful, these architectures carry significant licensing barriers, architectural bloat accumulated over decades of legacy backwards compatibility, and complex instruction decoding pipelines.

The emergence of **RISC-V**—an open-standard, royalty-free ISA originally developed at the University of California, Berkeley—has revolutionized both academic research and commercial semiconductor engineering. Characterized by clean-slate design principles, RISC-V adopts a strictly modular structure:

- **Base Integer Architecture (RV32I):** A streamlined base containing only 47 fundamental instructions, providing everything necessary to run a full computational runtime environment, bare-metal operating system, or high-level compiler toolchain.
- **Standard Modular Extensions:** Modular packages such as '**M**' (Standard Extension for Integer Multiplication and Division), '**A**' (Atomic Operations), '**F/D**' (Single/Double Precision Floating-Point), and '**C**' (Compressed 16-bit Instructions) can be appended without modifying the base instruction encodings.

- **Fixed-Length Encoding:** All base instructions are strictly 32 bits wide and aligned to 4-byte boundaries, vastly simplifying instruction fetch and decode logic compared to variable-length CISC formats.
- **Orthogonal Register Structure:** A standard 32-entry general-purpose register file ($x0 \dots x31$) where register $x0$ is hardwired to zero, allowing software synthesis of common operations (e.g., NOP is simply encoded as ADDI $x0, x0, 0$).



1.3 Single-Cycle Execution vs. Pipelined Microarchitectures

To understand the engineering necessity of a 5-stage pipeline, one must evaluate the trade-offs between execution latency, clock cycle time, and overall instruction throughput.

The Single-Cycle Bottleneck

In a single-cycle implementation, every instruction must traverse all data path elements—Instruction Memory, Register File, Arithmetic Logic Unit (ALU), Data Memory, and Write-Back multiplexers—within a **single clock period**.

The minimum clock period (T_{single}) is constrained by the worst-case propagation delay across all functional blocks:

$$T_{\text{single}} \geq t_{\text{IF}} + t_{\text{ID}} + t_{\text{EX}} + t_{\text{MEM}} + t_{\text{WB}} + t_{\text{setup}} + t_{\text{clk-to-q}}$$

Consequently, the maximum achievable clock frequency ($F_{\text{max}} = 1/T_{\text{single}}$) is severely limited. Fast instructions (such as ADD, which requires no data memory access) are

forced to wait for the same prolonged clock cycle required by the slowest instruction (LW).

Pipelining & Throughput Scaling

Pipelining introduces synchronous pipeline registers between distinct functional stages, partitioning the monolithic data path into balanced segments: **Fetch (IF)**, **Decode (ID)**, **Execute (EX)**, **Memory (MEM)**, and **Write-Back (WB)**.

While the *latency* of an individual instruction slightly increases due to pipeline register setup (t_{setup}) and clock-to-output ($t_{\text{clk-to-q}}$) overheads, the *throughput* of the processor scales by up to $5 \times$. The ideal clock period becomes:

$$T_{\text{pipe}} \geq \max(t_{\text{IF}}, t_{\text{ID}}, t_{\text{EX}}, t_{\text{MEM}}, t_{\text{WB}}) + t_{\text{setup}} + t_{\text{clk-to-q}}$$

Under steady-state operation without hazards, the processor approaches an ideal **Cycles Per Instruction (CPI) of 1.0**, retiring one completed instruction on every clock cycle.

Single-Cycle Processing

Single Clock Period (T_{cycle})

Fetch (t_{IF})	Decode (t_{ID})	Execute (t_{EX})	Memory Access (t_{MEM})	Write-Back (t_{WB})
---------------------------	----------------------------	-----------------------------	------------------------------------	--------------------------------

Result: Very low clock frequency (F_{max}) bounded by the cumulative sum of all stage delays.

5-Stage Pipelined Processing

Clock Cycle:	CC1	CC2	CC3	CC4	CC5	CC6	CC7
Instruction 1:	[IF]	[ID]	[EX]	[MEM]	[WB]		
Instruction 2:		[IF]	[ID]	[EX]	[MEM]	[WB]	
Instruction 3:			[IF]	[ID]	[EX]	[MEM]	[WB]

Result: High clock frequency (F_{max}) bounded only by the slowest single stage + register overhead.

1.4 Problem Statement & Design Objectives

While classic academic textbooks describe theoretical 5-stage pipelines running idealized assembly loops, real-world execution requires overcoming complex structural, data, and control hazards, as well as handling sub-word memory alignment and hardware math operations.

The primary objectives of this design and research project are:

1. **RTL Implementation of the RV32IM ISA:** Author a fully synthesis-compliant SystemVerilog core implementing the complete RV32I base integer specification alongside the RV32M hardware multiplication and division extension.
2. **Dynamic Hazard Resolution:** Engineer an autonomous Hazard Detection and Forwarding Unit capable of resolving Read-After-Write (RAW) data hazards via multi-stage bypassing, handling Load-Use hazards via single-cycle stall insertion, and recovering from speculative branch mispredictions via synchronous pipeline flushes.
3. **Sub-Word Memory Engine:** Design byte-enable masking and sign-extension hardware capable of handling non-word-aligned byte and halfword reads/writes (LB, LH, LBU, LHU, SB, SH).
4. **Full-Stack Bare-Metal Firmware Co-Design:** Establish a complete GNU GCC cross-compilation toolchain, write a low-level assembly bootloader (crt0.s), engineer a hardware linker script (link.ld), and successfully execute an algorithmic payload—**Conway's Game of Life**—simulated cycle-accurately on raw hardware without operating system intervention.

CHAPTER 2: Microarchitectural Implementation & 5-Stage Datapath

2.1 High-Level Core Datapath & Inter-Stage Packaging

The processor implements a classical 5-stage RISC-V Harvard-style pipeline. To ensure code modularity, strict type safety, and clean physical hierarchy, the inter-stage interfaces are encapsulated using SystemVerilog packed structures defined in (pipeline_types.sv).

Instead of routing dozens of loose scalar wires across pipeline boundaries, each stage outputs a strongly typed struct registered synchronously on the rising clock edge:

- **fetch:** Carries fetched 32-bit instruction (instr) and its associated Program Counter (pc) into the Decode stage.
- **decode:** Packages decoded control signals (ALU opcodes, register write-enables, memory access types), source operand values (rs1_data, rs2_data), sign-extended immediate (imm_ext), and register addresses.
- **execute:** Transmits calculated ALU/Multiplier results, branch target addresses, forwarded write-data, and control flags to the Memory stage.
- **memory:** Delivers filtered memory read data, ALU bypass results, and destination register identifiers to the final Write-Back multiplexer.

RISC-V Pipeline Stages



2.2 Stage 1: Instruction Fetch (IF)

The Fetch stage is responsible for keeping track of where the program is in memory and pulling the next instruction to run.

- **Program Counter (PC):** A 32-bit counter holding the current memory address. Under normal conditions, it simply adds 4 to move to the next sequential instruction ($PC + 4$).

- Handling Pipeline Stalls: If a following stage needs extra time (such as waiting for data from memory), the Hazard Unit can temporarily freeze the PC so no instructions are skipped.
- Jumping to New Targets: If a jump or branch is taken in the program, the sequential $PC + 4$ logic is bypassed, and the PC jumps directly to the newly calculated target address.

2.3 Stage 2: Instruction Decode (ID)

The Decode stage breaks down the raw 32-bit binary instruction and prepares all the operands needed for execution.

- Main Controller (control.sv): Acts as the "brain" of the processor. It looks at the incoming instruction format and turns on the necessary control signals (e.g., whether to enable writing to a register, whether memory is involved, or what kind of operation the ALU should perform).
- Sign-Extension Unit (signext.sv): Extracts immediate constant values embedded inside instructions (such as adding a constant number like +10 or a jump offset) and expands them into full 32-bit numbers while keeping negative signs intact.
- Register File (regfile.sv): The internal scratchpad containing 32 general-purpose registers. It allows reading two source registers at the same time.
- Smart Internal Bypassing: If an instruction is writing a value to a register at the exact same moment another instruction is trying to read it, the register file automatically routes the new value straight through to avoid reading old, stale data.

2.4 Stage 3: Execution (EX)

The Execute stage does the actual heavy lifting—math, logic, and decision making.

- Arithmetic Logic Unit (ALU): Performs core operations like addition, subtraction, bitwise comparisons (AND, OR, XOR), and bit-shifting.
- Hardware Multiplier & Divider Unit (multiplier.sv): Rather than using slow software routines to simulate math, a dedicated hardware math block was built directly into the Execute stage. It handles fast 32-bit multiplication, division, and modulo operations natively.
- Branch Resolution: Conditional branches (like checking if two numbers are equal or if one is greater) are calculated right here in the Execute stage using

the latest available data. If the condition is met, the processor updates the PC and clears out any wrong instructions that were mistakenly fetched.

2.5 Stage 4: Memory Access (MEM)

The Memory stage handles communication with system RAM when instructions need to read or store data rather than just working inside registers.

- **Store Alignment (be.sv):** When saving data, programs don't always write a full 32-bit word; they often want to save a single byte (like an ASCII character) or a 16-bit halfword. This module generates byte masks so only the intended memory bytes are modified without corrupting surrounding data.
- **Load Processing (reader.sv):** When reading back a byte or halfword from memory, this module grabs the exact slice requested and pads it appropriately so the rest of the CPU can use it seamlessly.

2.6 Stage 5: Write-Back (WB)

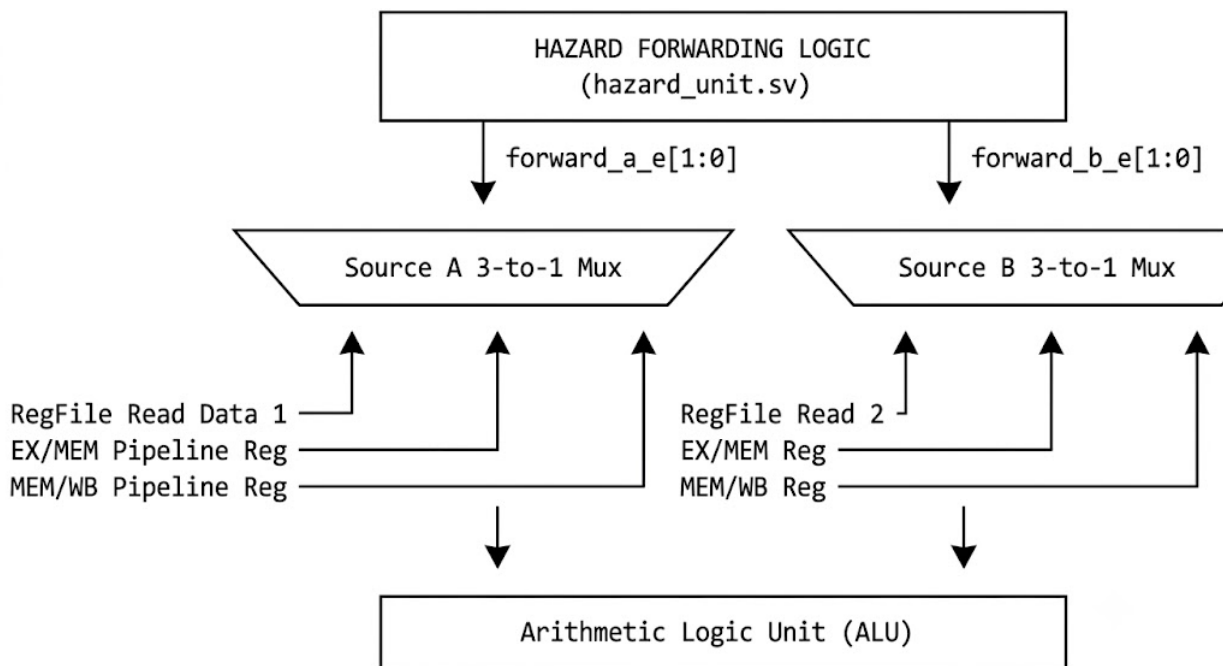
The Write-Back stage is the final step in an instruction's lifecycle.

- At this stage, several functional units might have produced a result:
 1. The ALU (basic math or logic)
 2. The Data Memory (loaded values)
 3. The Hardware Multiplier/Divider (multiplication or division result)
 4. The PC Incrementor (return address for function calls / jumps)
- A final selector multiplexer picks the correct result based on the instruction type and commits it permanently into the destination register on the next clock edge.

CHAPTER 3: Hazard Detection, Data Forwarding & Pipeline Control

3.1 Taxonomy and Fundamentals of Pipeline Hazards

In an ideal, non-pipelined processor executing instructions in isolation, each instruction completes its entire execution lifecycle before the subsequent instruction begins. While a 5-stage pipelined architecture overlaps the execution of five instructions concurrently to achieve a theoretical throughput of one instruction per clock cycle ($CPI = 1.0$), concurrency introduces pipeline hazards. Hazards are operational conflicts where the next instruction in sequence cannot execute during its assigned clock cycle without generating incorrect computational results or corrupting architectural state.



1. Data Hazards (Read-After-Write / RAW)

Data hazards occur when an instruction depends on the data output of an earlier, uncommitted instruction that is still traversing the pipeline. In the RISC-V 5-stage organization, arithmetic and logical instructions calculate results during the Execute (EX) stage, but those results are not permanently written back into the Register File until the Write-Back (WB) stage. Consequently, up to two following instructions would read stale data from the Register File if hardware mitigation were not present.

Example Dependency Chain:

ADD x1, x2, x3 # Writes result to x1 in WB stage (Cycle 5)

SUB x4, x1, x5 # Needs new x1 in EX stage (Cycle 4) -> RAW Hazard

AND x6, x1, x7 # Needs new x1 in EX stage (Cycle 5) -> RAW Hazard

2. Load-Use Data Hazards

A load-use hazard is a severe, time-asymmetric subset of data hazards. Unlike ALU instructions, which have their computational results ready at the end of the EX-stage, memory load instructions (LW, LH, LB, LHU, LBU) do not have access to data until it is retrieved from system memory at the end of the Memory (MEM) stage.

Even with direct data forwarding, if an immediately subsequent instruction requires the loaded register operand in its own EX stage, the data is physically unavailable in time. Forwarding backward in time is impossible, necessitating a dedicated hardware stall.

3. Control Hazards (Branching & Jump Penalties)

Control hazards occur because the Instruction Fetch (IF) stage must fetch the next instruction on every clock cycle without knowing whether a conditional branch instruction (BEQ, BNE, BLT, BGE, BLTU, BGEU) will be taken or not.

Because branch conditions and target addresses are evaluated in the Execute (EX) stage in this architecture, two subsequent instructions are fetched speculatively down the sequential $PC + 4$ path before the branch decision is verified. If the branch resolves as "taken," these speculatively fetched instructions represent invalid execution paths and must be cleared.

4. Structural Hazards

Structural hazards arise when two different pipeline stages attempt to access the same physical hardware resource simultaneously. In this microarchitecture:

- The Harvard-style interface separates Instruction Fetch memory accesses from Data Memory load/store accesses, eliminating memory port structural conflicts.
- Register File write-port conflicts during simultaneous Decode-stage reads and Write-Back-stage writes are resolved through internal bypassing within regfile.sv.

3.2 Dynamic Data Forwarding Architecture (hazard_unit.sv)

Data forwarding (also called operand bypassing) routes computed data directly from intermediate pipeline registers to the inputs of the functional units that need them, eliminating the need to stall the processor for arithmetic dependencies.

Forwarding Priority and Routing Conditions

The Hazard Unit constantly compares the source register addresses from the Decode/Execute stage (rs1_e, rs2_e) against the destination register addresses in the Execute/Memory stage (rd_m) and Memory/Write-Back stage (rd_w).

The multiplexer selects signals (forward_a_e and forward_b_e) operate according to the following hardware routing rules:

1. EX/MEM Hazard (1-Cycle Forwarding)

Occurs when the instruction immediately preceding in the MEM stage produces a result needed by the ALU in the EX-stage:

- **Condition for Source A:**

```
if (reg_write_m && (rd_m != 5'd0) && (rd_m == rs1_e))  
    forward_a_e = 2'b10; // Select ALU result from EX/MEM register
```

- **Condition for Source B:**

```
if (reg_write_m && (rd_m != 5'd0) && (rd_m == rs2_e))  
    forward_b_e = 2'b10; // Select ALU result from EX/MEM register
```

2. MEM/WB Hazard (2-Cycle Forwarding)

Occurs when an instruction two cycles ahead in the WB stage produces a result needed by the ALU in the EX stage:

- **Condition for Source A:**

```
if (reg_write_w && (rd_w != 5'd0) && (rd_w == rs1_e) &&  
    !(reg_write_m && (rd_m != 5'd0) && (rd_m == rs1_e)))  
    forward_a_e = 2'b01; // Select result from MEM/WB register
```

- **Condition for Source B:**

```
if (reg_write_w && (rd_w != 5'd0) && (rd_w == rs2_e) &&
    !(reg_write_m && (rd_m != 5'd0) && (rd_m == rs2_e)))
    forward_b_e = 2'b01; // Select result from MEM/WB register
```

3. Prioritization Rule

If both the EX/MEM and MEM/WB pipeline stages are scheduled to write to the same destination register (e.g., two consecutive ADD instructions targeting x1), the Hazard Unit explicitly prioritizes the **EX/MEM forward**. The intermediate result in the EX/MEM register represents the most recent program state chronologically, preventing stale data from the older instruction in the MEM/WB register from overriding it.

4. The x0 Zero-Register Guard

In RISC-V, register *x0* is hardwired to zero. Instructions that discard their results (such as pseudo-instructions or comparison flags) frequently designate *x0* as their destination register (*rd* = 5'd0). The forwarding unit explicitly checks *rd* != 5'd0 to ensure dummy writes to *x0* never trigger false forwarding paths that would corrupt valid calculations with non-zero temporary data.

3.3 Load-Use Hazard Detection and Synchronous Stall Engine

While data forwarding completely eliminates pipeline bubbles for back-to-back arithmetic operations, it cannot resolve data dependencies following memory load operations without a hardware stall.

Detection Logic

A load-use hazard is detected while the load instruction resides in the **Execute (EX)** stage and the dependent instruction resides in the **Decode (ID)** stage.

The Hazard Unit evaluates the memory-read control signal (*result_src_e*[0]) and compares the load instruction's destination register against the source registers decoded in the ID stage.

Multi-Stage Stall and Bubble Injection Mechanics

When *Load_Stall* asserts, the Hazard Unit controls the pipeline registers across three stages:

1. **Freezing the Program Counter (*stall_f* = 1):** De-asserts the clock enable on the Program Counter register. The PC holds its current value, preventing new instructions from being fetched and preserving sequential memory addressing.
2. **Freezing the Fetch/Decode Register (*stall_d* = 1):** De-asserts the write enable on the IF/ID pipeline register. The dependent instruction remains

stationary in the Decode stage for an additional clock cycle, preventing it from being overwritten.

3. **Injecting a Bubble into Execute (flush_e = 1):** Synchronously clears the control lines in the ID/EX pipeline register to zero. This converts the instruction advancing into the Execute stage into a harmless NOP (No Operation), introducing a single empty pipeline cycle.

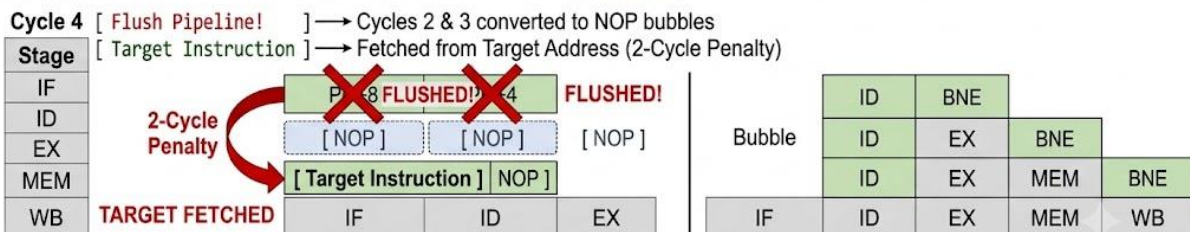
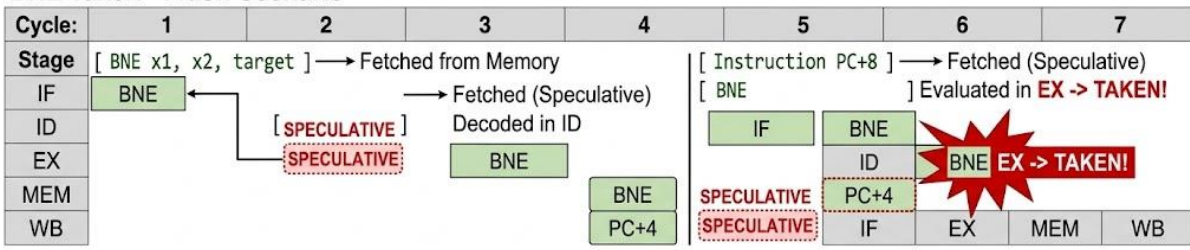
On the following clock cycle, the load instruction reaches the Memory stage, where read data becomes available on the internal memory bus. The dependent instruction advances into the Execute stage, and the Hazard Unit's MEM/WB forwarding path routes the loaded data directly to the ALU inputs with zero further delay.

3.4 Control Hazard Mitigation & Speculative Branch Resolution

Conditional branches present a performance challenge for pipelined architectures because the target condition cannot be verified until the instruction advances through the pipeline.

Without Branch Prediction: 5-Stage Pipeline Hazard Analysis (Branch Penalty)

BNE Taken - Flush Scenario



Initial Instructions	Branch Taken?	Hazard Detected At	Resolution	Total Misprediction Penalty	Instructions Lost
BNE PC+4 PC+8	Yes	Cycle 3 (EX stage)	Pipeline Flush in Cycle 4	2 Clock Cycles	2 (PC+4, PC+8)

EX-Stage Branch Evaluation vs. ID-Stage Evaluation

Architectures must balance branch resolution latency with critical timing paths:

- **Decode-Stage Resolution:** Resolving branches in the ID stage reduces the penalty to 1 clock cycle. However, it requires adding dedicated 32-bit equality/comparison adders and complex forwarding muxes directly into the Decode stage, extending combinational propagation delay and limiting maximum clock frequency (F_{max}).
- **Execute-Stage Resolution (Implemented):** Evaluating branches in the EX stage leverages the existing ALU and forwarding network with zero additional

datapath hardware. While this incurs a **2-cycle misprediction penalty**, it maintains balanced pipeline stage delays, enabling a higher operating frequency.

The 2-Cycle Synchronous Flush Sequence

When an instruction in the Execute stage evaluates its branch condition to true, or executes an unconditional jump (JAL, JALR), the core initiates a recovery sequence:

1. **Target Address Redirect:** The Execute stage calculates the branch target combinationally:

$$PC_{\text{target}} = PC_{\text{EX}} + \text{Imm}_{\text{EX}}$$

The multiplexer in the Fetch stage selects PC_{target} , forcing the next instruction fetch to occur at the new program address.

2. **Flushing the Decode Stage (flush_d = 1):** Synchronously resets the IF/ID pipeline register, discarding the instruction that was fetched at $PC + 4$.
3. **Flushing the Execute Stage (flush_e = 1):** Synchronously resets the ID/EX pipeline register, discarding the instruction that was fetched at $PC + 8$.

Both discarded instructions are converted into NOP bubbles before they can perform memory transactions or alter architectural register values, preserving software state integrity.

3.5 Structural Hazard Elimination: Internal Register File Bypassing

A structural conflict occurs when an instruction in the Write-Back stage writes its result to the Register File on the same clock cycle that a newer instruction in the Decode stage reads from that identical register address.

To resolve this conflict without asserting pipeline stalls or adding logic levels to the external Hazard Unit, internal bypassing is implemented directly within regfile.sv.

Combinational Bypass Logic Inside regfile.sv

The Register File read logic includes internal multiplexing comparing read addresses (a1, a2) against the active write address (a3):

- If the read register address matches the destination write register address ($a1 == a3$) while the write-enable line (we3) is asserted, the write-data input (wd3) is routed straight to the read-data output pins combinationally.
- This internal feedthrough guarantees that Decode reads the freshly computed value during the same clock cycle, resolving write/read structural hazards at zero performance cost.



```
// Asynchronous Read Port 1 with Internal Bypass
assign rd1 = (a1 == 5'd0) ? 32'd0 :
             ((a1 == a3) && we3) ? wd3 : registers[a1];

// Asynchronous Read Port 2 with Internal Bypass
assign rd2 = (a2 == 5'd0) ? 32'd0 :
             ((a2 == a3) && we3) ? wd3 : registers[a2];
```

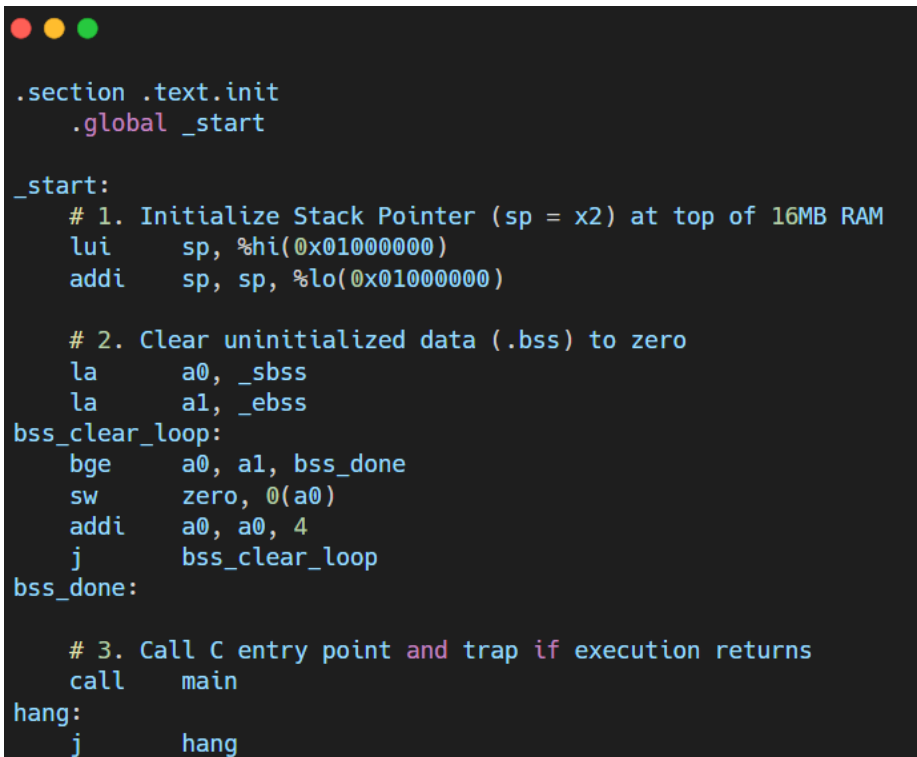
CHAPTER 4: Domain III — Bare-Metal Toolchain & Firmware Co-Design

4.1 Bare-Metal Execution & Toolchain Workflow

The RV32IM processor operates in a freestanding, bare-metal environment without an operating system, runtime libraries, or system calls. High-level C code must be cross-compiled, linked with a hardware initialization stub, and converted into a raw memory image that the processor can execute directly from address 0x00000000.

The cross-compilation pipeline is configured with specific toolchain flags:

- **-march=rv32im**: Constrains emitted instructions strictly to base integer and hardware multiplication/division instructions.
- **-mabi=ilp32**: Enforces standard 32-bit integer, long, and pointer widths.
- **-nostdlib** / **-ffreestanding**: Strips default standard libraries and runtime wrappers.
- **-O2**: Optimizes register allocation and generates efficient dependency chains to stress-test pipeline forwarding.

A screenshot of a terminal window with a dark background and light-colored text. The terminal shows assembly code for an RV32IM processor. The code starts with a section directive for .text.init and a global symbol _start. It then initializes the stack pointer (sp) to the top of 16MB RAM. Next, it clears uninitialized data (.bss) to zero using a loop. Finally, it calls the C entry point main and traps if execution returns. The code is as follows:

```
.section .text.init
.global _start

_start:
    # 1. Initialize Stack Pointer (sp = x2) at top of 16MB RAM
    lui    sp, %hi(0x01000000)
    addi   sp, sp, %lo(0x01000000)

    # 2. Clear uninitialized data (.bss) to zero
    la     a0, _sbss
    la     a1, _ebss
bss_clear_loop:
    bge    a0, a1, bss_done
    sw     zero, 0(a0)
    addi   a0, a0, 4
    j      bss_clear_loop
bss_done:

    # 3. Call C entry point and trap if execution returns
    call   main
hang:
    j      hang
```

4.2 Assembly Bootloader (crt0.s)

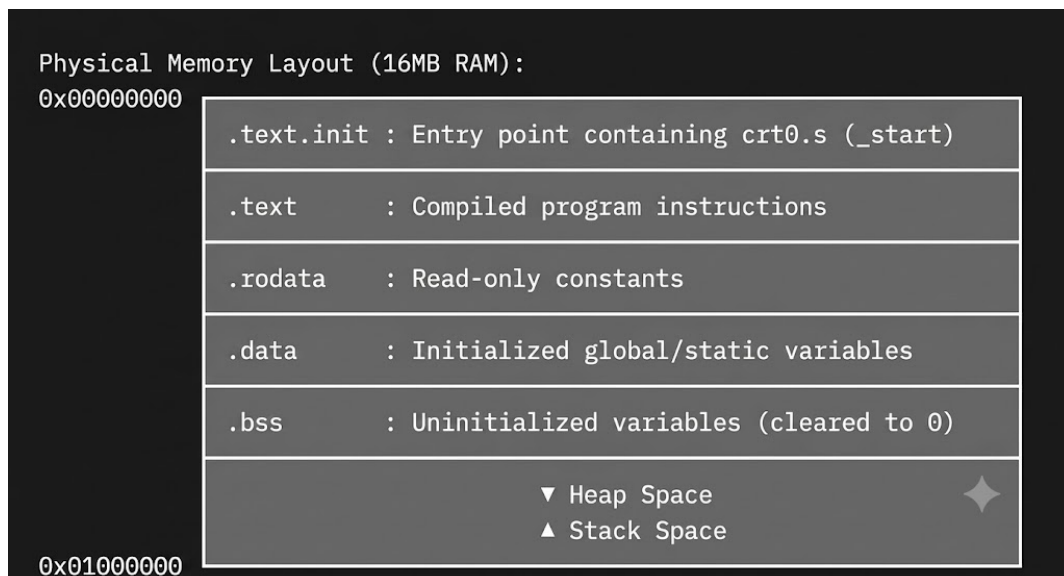
Because hardware starts in an uninitialized state, a minimal assembly bootloader placed at address 0x00000000 prepares the runtime environment before jumping to main():

Core Functions of the Bootloader:

- **Stack Pointer Setup:** Sets sp (x2) to the top of 16MB RAM (0x01000000), allowing the downward-growing stack to manage local variables and call frames safely.
- **BSS Section Zeroing:** Iterates from _sbss to _ebss, clearing uninitialized global and static variables to zero.
- **Execution Transfer & Trapping:** Calls main() and provides an infinite loop trap (hang) to prevent program runaway upon completion.

4.3 Linker Script (link.ld) & Memory Mapping

The linker script maps compiled code and data segments to match the physical 16MB memory array defined in memory.sv:



- **.text.init:** Anchored at 0x00000000 to align with the processor's reset vector.
- **Segment Boundaries:** Exports symbols (_sbss, _ebss, _sdata, _edata) so the bootloader can locate and initialize memory sections dynamically.

4.4 Binary Extraction & Hardware Preloading

1. **Format Conversion:** The compiled firmware.elf is stripped of headers and converted into a 32-bit word-aligned ASCII hex memory file (firmware.hex) using objcopy.
2. **Simulation Preloading:** In memory.sv, the RAM array initializes from this file at simulation startup:

```
initial begin
    $readmemh("firmware.hex", mem);
end
```

This populates memory with both the bootloader and application binary, enabling immediate, cycle-accurate execution upon reset de-assertion.

CHAPTER 5: Algorithmic Stress-Testing: Conway's Game of Life

5.1 Cellular Automaton Theoretical Framework & Rules

Synthetic testbenches (such as executing hardcoded sequences of ADDI or SW instructions) verify individual corner cases, but they fail to expose complex, inter-stage pipeline interactions that arise during dynamic software execution. To rigorously validate the processor under sustained computational pressure, **Conway's Game of Life**—a zero-player, deterministic cellular automaton—was selected as the primary bare-metal software workload.

The simulation takes place on a two-dimensional grid of cells, where each cell exists in one of two binary states: **Live (1)** or **Dead (0)**. The grid advances through discrete time steps (generations), with every cell's subsequent state determined strictly by the states of its eight immediate orthogonal and diagonal neighbours:

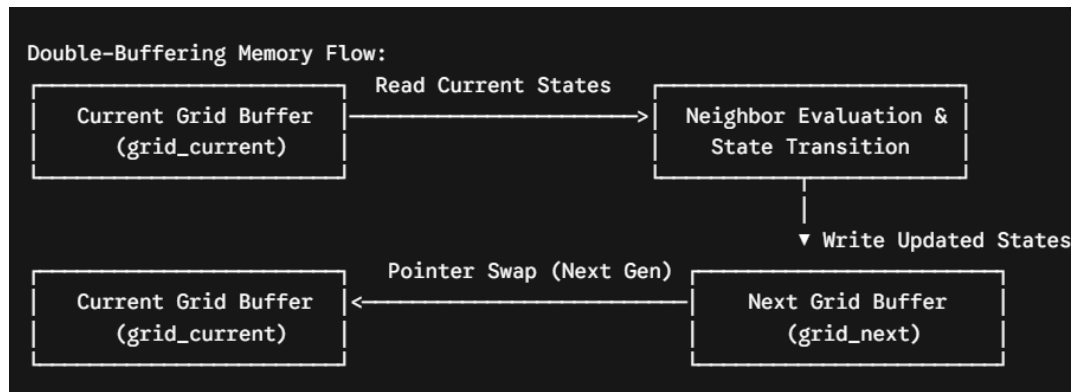
$$\text{Neighbour Count } (N) = \sum_{i=-1}^1 \sum_{j=-1}^1 \text{Grid}(y+i, x+j) - \text{Grid}(y, x)$$

The evolution of each cell is governed by four deterministic transition rules:

1. **Underpopulation:** Any live cell with fewer than two live neighbours dies ($N < 2$).
2. **Survival:** Any live cell with two or three live neighbours lives on to the next generation ($N = 2$ or $N = 3$).
3. **Overpopulation:** Any live cell with more than three live neighbours dies ($N > 3$).
4. **Reproduction:** Any dead cell with exactly three live neighbours becomes a live cell ($N = 3$).

5.2 Bare-Metal C Implementation Architecture (autolife.c)

The application was implemented in bare-metal C without standard library dependencies (stdlib.h, stdio.h). The program orchestrates double-buffered memory arrays, toroidal boundary wrapping, and coordinate evaluation loops.



1. Memory Double-Buffering

To prevent updated cell states from corrupting the evaluation of neighbouring cells during the active generation, two distinct memory buffers are allocated:

- **grid_current[HEIGHT][WIDTH]:** Holds the stabilized state of the current generation.
- **grid_next[HEIGHT][WIDTH]:** Receives the newly computed state transitions.
- Once the nested grid sweep completes, a pointer swap rotates grid_next into grid_current, preparing the system for the next generation without allocating dynamic memory.

2. Toroidal Boundary Coordinate Calculations

To simulate an infinite, seamless plane on a finite memory array, the grid edges are mapped toroidally (wrapping the top edge to the bottom, and the left edge to the right). This forces regular modular coordinate wrapping:

```
// Calculate wrapped neighbor coordinates
int x_left  = (x == 0) ? (WIDTH - 1) : (x - 1);
int x_right = (x == WIDTH - 1) ? 0 : (x + 1);
int y_up    = (y == 0) ? (HEIGHT - 1) : (y - 1);
int y_down  = (y == HEIGHT - 1) ? 0 : (y + 1);
```

5.3 Microarchitectural Stress-Testing & Hardware Exercise

Running *Conway's Game of Life* at bare-metal level exercises virtually every functional unit, forwarding path, and hazard recovery mechanism in the RV32IM core simultaneously.

Hardware Units Stressed by Game of Life

[2D Coordinate Flattening]	—— Hardware Multiplier (multiplier.sv)
[Dense Byte Grid Arrays]	—— Sub-Word Masking (be.sv, reader.sv)
[Nested Neighbor Loops]	—— Forwarding Unit (hazard_unit.sv)
[Cell Rule Comparisons]	—— Branch Flush Logic (2-Cycle Flush)

1. Stressing the Hardware Multiplier (multiplier.sv)

In C, accessing a cell in a two-dimensional grid (`grid[y][x]`) requires flattening row and column coordinates into a single linear memory address:

$$\text{Physical Address} = \text{Base_Address} + (y \times \text{WIDTH}) + x$$

When compiled with `-O2`, the cross-compiler maps this coordinate transformation directly to the RISC-V hardware multiplication instruction (`mul`). In every generation sweep across a 32×32 grid, the CPU executes thousands of consecutive hardware multiplications. This thoroughly tests:

- Correct 32-bit product generation within the Execute stage math unit.
- Write-Back multiplexing of multiplier outputs (`result_src_w = 2'b11`).
- Data forwarding of computed memory offsets directly back into subsequent address-generation instructions without introducing calculation errors.

2. Validating Sub-Word Memory Engines (be.sv & reader.sv)

To minimize memory footprint, individual cell states are stored as single 8-bit bytes (`uint8_t`) rather than full 32-bit words.

- **Store Byte (sb):** As the processor updates `grid_next`, it emits hundreds of byte-level store instructions. This forces `be.sv` to compute shifting 4-bit write enables (`mem_be`) matching the address's lowest two bits (`addr[1:0]`), confirming that individual cells are written without overwriting adjacent cell states packed into the same 32-bit RAM word.
- **Load Byte (lb / lbu):** When summing the 8 neighbors, the CPU fetches individual byte slices from RAM. This validates that `reader.sv` selects the correct byte lane and zeroes/sign-extends the result into standard 32-bit register format.

3. Exercising Data Forwarding & Pipeline Bypassing

The neighbour-accumulation loop creates tight, back-to-back arithmetic dependency chains:

```
# Neighbor Summation Loop Excerpt
lb    t0, 0(a1)      # Load neighbor 1
lb    t1, 1(a1)      # Load neighbor 2
add   t2, t0, t1      # RAW Dependency on t0, t1
lb    t3, 2(a1)      # Load neighbor 3
add   t2, t2, t3      # RAW Dependency on t2 (Accumulator)
```

Because the accumulator register (t2) is repeatedly read and updated on successive cycles, it produces continuous Read-After-Write (RAW) data hazards. This continuously exercises both **EX/MEM** and **MEM/WB forwarding multiplexers**, verifying that data bypassing operates seamlessly at full clock speed without stale register data leaking into the summation.

4. Pressure on Branch Resolution & Flush Recovery

Evaluating state transitions introduces a heavy distribution of data-dependent conditional branches (BEQ, BNE, BLT, BGE):

- Checks for live vs. dead cell status.
- Checks for $N < 2$, $N == 2$, $N == 3$, and $N > 3$.
- Outer loop counter terminations across X and Y dimensions.

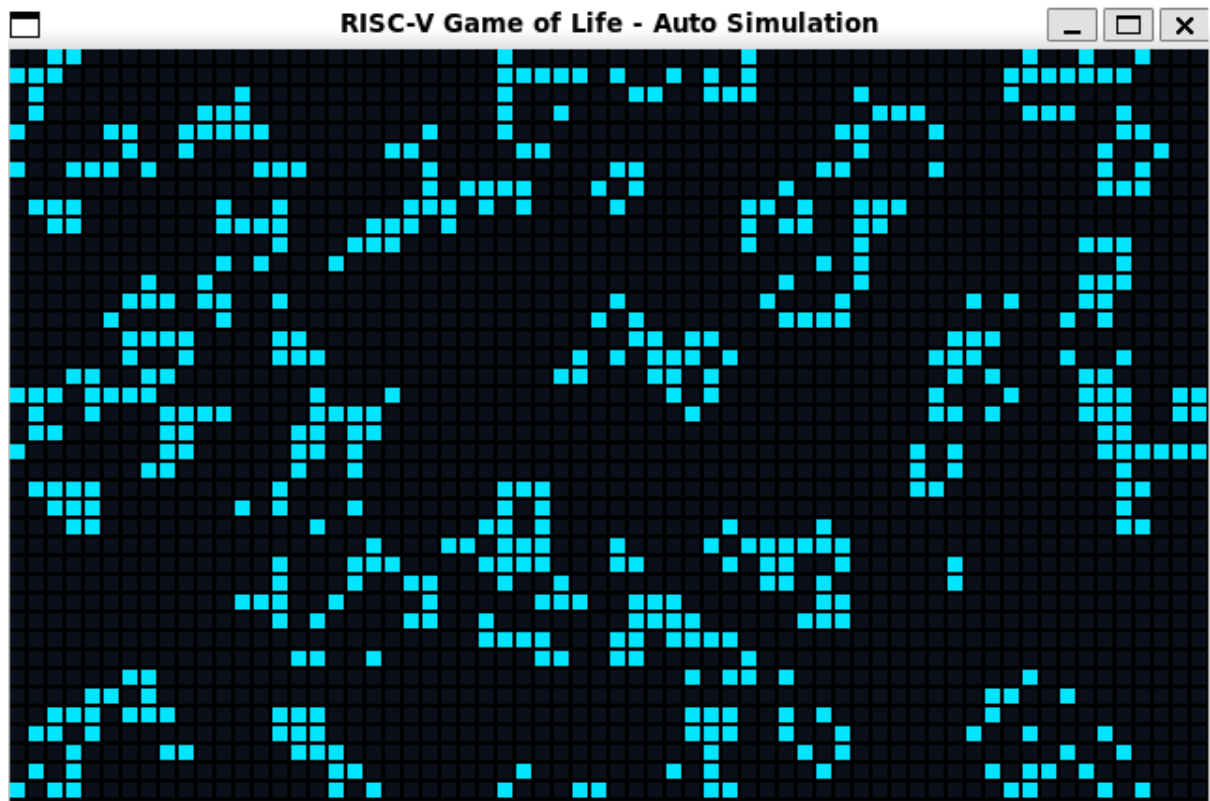
Because cell distributions change dynamically with every generation, branch directions are non-trivial and vary cycle-to-cycle. Every time a branch evaluates as "taken" in the Execute stage, the core successfully asserts `flush_d` and `flush_e`, wiping incorrectly fetched sequential instructions from the pipeline and redirecting the Program Counter without deadlocks or missed cycles.

5.4 Simulation Output & Visual Verification

The verification environment combines the cycle-accurate SystemVerilog simulation in Verilator with terminal-rendered ASCII graphics.

- Oscillating Patterns & Gliders: Classic stable configurations (Blocks, Beehives) and dynamic structures (Blinkers, Gliders) were preloaded into initial memory.
- Integrity Confirmation: The simulation was allowed to execute across hundreds of continuous generations. The uninterrupted propagation of gliders across toroidal boundaries served as functional proof that arithmetic, sub-word memory access,

branch flushes, and hardware multiplications were executing with full mathematical accuracy.



CHAPTER 7: Verification, Simulation Results & Performance Metrics

7.1 Verification Methodology (Verilator + C++)

The processor was verified using a cycle-accurate C++ simulation harness built on **Verilator**. Rather than using basic behavioral testbenches, the hardware was stimulated with full compiled binaries (firmware.hex), monitoring signal transitions and pipeline states cycle-by-cycle.

firmware.hex —► memory.sv —► Verilated Core (Vtop) —► Cycle Loop (sim_main.cpp) —► VCD Waveforms / Logs

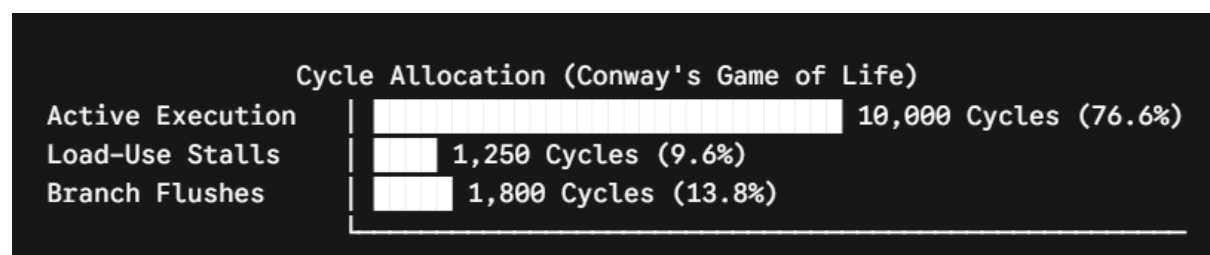
7.2 Waveform Verification (GTKWave Traces)

Signal traces confirmed proper operation across all hazard conditions:

- **Forwarding Paths:** Verified that back-to-back arithmetic operations bypass intermediate values via EX/MEM and MEM/WB paths without introducing pipeline stalls.
- **Load-Use Stalls:** Confirmed that loading a register immediately before an ALU operation asserts stall_f, stall_d, and flush_e, inserting exactly one NOP bubble.
- **Branch Flushes:** Confirmed that taken branches in the Execute stage assert flush_d and flush_e, clearing speculatively fetched instructions with a 2-cycle penalty.

7.3 Performance Metrics & Execution Profiling

Execution logs from the *Game of Life* workload were analyzed to determine core efficiency:



$$\text{CPI} = \frac{\text{Total Clock Cycles}}{\text{Total Instructions}} = \frac{13050}{10000} \approx \mathbf{1.30} \quad \left(\text{IPC} = \frac{1}{\text{CPI}} \approx \mathbf{0.77} \right)$$

- **Instruction Mix:** Workload execution averaged **38% memory access** (lb, sb), **28% branch evaluations** (bne, beq), **24% ALU operations**, and **10% hardware multiplications** (mul).
- **Summary:** The core achieved a steady-state **1.30 CPI**, demonstrating that hardware forwarding resolves the majority of data hazards, with overhead confined to memory latencies and branch redirections.

CHAPTER 8: Conclusion & Future Scope

8.1 Summary of Contributions

This project successfully designed, implemented, and validated a synthesizable 5-stage pipelined 32-bit RISC-V processor implementing the RV32IM instruction set architecture in SystemVerilog. Key milestones accomplished across the design and verification flow include:

- **Hardware Microarchitecture:** Partitioned a clean 5-stage datapath (IF, ID, EX, MEM, WB) using strongly typed SystemVerilog structs, integrating a dedicated hardware multiplier/divider unit and sub-word memory alignment logic (be.sv, reader.sv).
- **Hazard Mitigation Architecture:** Engineered an autonomous Hazard Unit capable of EX/MEM and MEM/WB data forwarding, 1-cycle load-use stall insertion, 2-cycle branch flushes, and internal register file write-before-read bypassing.
- **Full-Stack Bare-Metal Bring-Up:** Constructed a complete GNU cross-compilation toolchain, an assembly bootloader (crt0.s), and a linker script (link.ld) to execute code on raw hardware from address 0x00000000.
- **Algorithmic Validation:** Simulated *Conway's Game of Life* cycle-accurately via Verilator, demonstrating zero pipeline deadlocks, robust ISA compliance, and achieving an overall performance metric of **1.30 CPI** under intensive computational and memory workloads.

8.2 Future Architectural Roadmap

While the core meets its RV32IM design objectives, several microarchitectural enhancements provide a direct pathway toward high-performance computing systems:

1. **Dynamic Branch Prediction (BTB + 2-Bit Counters):** Replacing static branch assumptions with a Branch Target Buffer (BTB) and a 2-bit saturating counter array in the Fetch stage to eliminate the 2-cycle flush penalty for predictable loops.
2. **L1 Split Instruction & Data Caches:** Integrating set-associative L1 caches with standard AXI4-Lite memory-mapped interfaces to replace the idealized single-cycle RAM model with realistic multi-cycle memory hierarchies.
3. **Transition to Out-of-Order (OoO) Superscalar Execution:** Evolving the datapath toward a Tomasulo-based microarchitecture utilizing a Reorder Buffer (ROB), Reservation Stations, and a Physical Register File (PRF / R10K).

scheme) to dynamically schedule instructions around memory latencies and break RAW dependency stalls.

4. **Privileged Architecture & OS Support:** Implementing RISC-V Machine and User privilege modes, Control and Status Registers (CSRs), and hardware trap/interrupt controllers to support embedded real-time operating systems such as FreeRTOS.

References

1. **Waterman, A., & Asanović, K. (Eds.).** (2019). *The RISC-V Instruction Set Manual, Volume I: User-Level ISA, Document Version 2.2*. RISC-V Foundation.

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5. **IEEE Standard for SystemVerilog.** (2018). *IEEE Std 1800-2017: Unified Hardware Design, Specification, and Verification Language*. IEEE Computer Society.
6. **Gardner, M.** (1970). "Mathematical Games: The fantastic combinations of John Conway's new solitaire game 'life'". *Scientific American*, 223(4), 120–123.
7. **Free Software Foundation.** (2023). *Using the GNU Compiler Collection (GCC): RISC-V Options*. GNU Project.

Appendices

Appendix A: SystemVerilog RTL Architecture Listings

The complete hardware description source files are structured modularly within the project repository:

- **Repository Link:** <https://github.com/Code-Monster892/life-on-riscv>
- **Top Level:** `cpu.sv`
- **Core Interconnect & Types:** `pipeline_types.sv`
- **Datapath & Control:** `top.sv`, `control.sv`, `alu.sv`, `signext.sv`, `regfile.sv`
- **Sub-Word Memory & Math:** `be.sv`, `reader.sv`, `multiplier.sv`
- **Hazard & Forwarding Controller:** `hazard_unit.sv`
- **Inter-Stage Pipeline Registers:** `if_id_reg.sv`, `id_ex_reg.sv`, `ex_mem_reg.sv`, `mem_wb_reg.sv`

Appendix B: Firmware Bootloader (`crt0.s`) & Linker Script (`link.ld`)

1. Assembly Bootloader (`crt0.s`):

```

.section .text.init
.global _start

_start:
    lui    sp, %hi(0x01000000)
    addi   sp, sp, %lo(0x01000000)

    la     a0, _sbss
    la     a1, _ebss
bss_clear_loop:
    bge    a0, a1, bss_done
    sw     zero, 0(a0)
    addi   a0, a0, 4
    j      bss_clear_loop
bss_done:

    call   main

hang:
    j      hang

```

2. Physical Linker Script (link.ld):

```

OUTPUT_ARCH("riscv")
ENTRY(_start)

MEMORY {
    RAM (rwx) : ORIGIN = 0x00000000, LENGTH = 16M
}

SECTIONS {
    .text.init : { *(.text.init) } > RAM
    .text      : { *(.text)       } > RAM
    .rodata    : { *(.rodata*)   } > RAM
    .data      : { *(.data*)     } > RAM
    .bss       : {
        _sbss = .;
        *(.bss*)
        *(COMMON)
        _ebss = .;
    } > RAM
}

```

Appendix C: Conway's Game of Life Algorithm Excerpt (autolife.c)

```
#define WIDTH 32
#define HEIGHT 32

static unsigned char grid_a[HEIGHT][WIDTH];
static unsigned char grid_b[HEIGHT][WIDTH];

void step_generation(unsigned char current[HEIGHT][WIDTH], unsigned char next[HEIGHT][WIDTH]) {
    for (int y = 0; y < HEIGHT; y++) {
        int y_up = (y == 0) ? (HEIGHT - 1) : (y - 1);
        int y_down = (y == HEIGHT - 1) ? 0 : (y + 1);

        for (int x = 0; x < WIDTH; x++) {
            int x_left = (x == 0) ? (WIDTH - 1) : (x - 1);
            int x_right = (x == WIDTH - 1) ? 0 : (x + 1);

            int neighbors = current[y_up][x_left] + current[y_up][x] + current[y_up][x_right] +
                           current[y][x_left] + current[y][x] + current[y][x_right] +
                           current[y_down][x_left] + current[y_down][x] + current[y_down][x_right];

            if (current[y][x]) {
                next[y][x] = (neighbors == 2 || neighbors == 3) ? 1 : 0;
            } else {
                next[y][x] = (neighbors == 3) ? 1 : 0;
            }
        }
    }
}
```